

Problem

Agent payments need privacy and auditability

- Autonomous agents pay each other, buy API access, and move micro-capital at high frequency.
- Publishing every amount on-chain leaks strategy and competitive structure.
- Staying fully off-chain removes proof for settlement, compliance, and ecosystem partners.
- **Why Circle / Arc:** Programmable USDC and sub-cent settlement are built for agentic commerce.
- Builders lack a reference stack: live agent economy + Circle rails + credible confidentiality.

SymbioMarket - open coordination, encrypted amounts, public settlement proof.

Solution

Live four-agent economy -> FHE ledger -> Arc settlement

- Oracle -> Strategist -> Executor -> Evaluator (cycle every ~6s)
- Python swarm_api.py -> swarm_data.json -> Next.js /swarm dashboard

Three layers

Layer	Where	Hidden	Public
Agent economy	Dashboard	-	Who paid whom, purpose, cycle ID
Confidential ledger	Sepolia + Zama FHE	Payment amounts	Contract + tx hash
Settlement pulse	Arc testnet	-	Wallet, tx, Arcscan proof

Flow: Swarm -> sync:swarm -> Sepolia FHECounter -> Arc pulse -> fhe_sync_state.json -> UI

Demo & on-chain proof

Shipped on testnet - not mock UI

- python3 agents/swarm_api.py - four agents, real cycles
- arc-nanopayments: npm run dev -> localhost:3000/swarm
- fhe-contracts: npm run sync:swarm - FHE sync + Arc pulse

Deployed contract (Sepolia)

FHECounter: `0x8Fe90e590E58b19127B760D07F4e79655bb90DEf`

Proof transactions

FHE: `sepolia.etherscan.io/tx/0xe3c7899085b72092c4c33504d8de71cc5ee0a6cd855767641fe6fcec8744bce6`

Arc: `testnet.arcscan.app/tx/0x8e3076272c2bcb0b2e431627098f8fda471ebf8ab38730362c2555a9f20260a8`

Traction: pre-production demo; 759+ swarm cycles in development. Grant: public deploy + USDC volume on Arc testnet.

Circle integrations

Built on Circle stack - extending, not greenfield

- Nanopayments / x402: arc-nanopayments fork; premium APIs; buyer/seller flow
- Circle Gateway: Balance + withdraw UI (app/api/gateway/)
- USDC: Arc testnet USDC in x402 config; documented testnet flows
- Arc: arc_settle_swarm.py - confirmed settlement txs on Arcscan

Grant video paths: swarm_api.py, lib/x402.ts, gateway APIs, sync-swarm-payment.ts

- Planned: full x402 USDC on Arc; Gateway withdraw; USDC settlement per payment

Milestones & team

12-week grant plan

M1 (wk 1-4)	Public demo + Nanopayments	Deployed /swarm; video; Arc USDC tx via Circle stack
M2 (wk 5-8)	Hardened swarm + Arc/cycle	100+ payments; 10+ Arc settlements linked
M3 (wk 9-12)	FHE treasury + mainnet plan	Extended FHE; builder playbook; repro <1 hr

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Built SymbioMarket for Arc hackathon: multi-agent swarm, Circle nanopayments, Zama FHE ledger, Arc testnet settlement. Nigeria | solo | not otherwise funded | open source.

Confidential agent economics on programmable USDC rails.